

## Square Roots — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

### Objective

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**Q1** (b) **25.**  $25 \times 25 = 625$ .

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**Q2** (a) **34.**  $1156 = 2^2 \times 17^2$ , so  $\sqrt{\phantom{x}} = 2 \times 17 = 34$ .

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**Q3** 7.  $2800 = 2^4 \times 5^2 \times 7$ ; only the 7 is unpaired, so multiply by 7.

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**Q4** (c) **11025.**  $11025 = 3^2 \times 5^2 \times 7^2$  (all paired); the others have unpaired primes.

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**Q5** **True.** e.g.  $\sqrt{64} = +8$  and  $-8$ .

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**Q6** (d) **A false, R true.**  $156 = 2^2 \times 3 \times 13$  has unpaired 3 and 13, so it is not a perfect square; the reason is a correct statement.

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### Short answer

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**Q7** **69.**  $4761 = 3^2 \times 23^2$ , so  $\sqrt{\phantom{x}} = 3 \times 23 = 69$ .

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**Q8**  **$15 < \sqrt{250} < 16$ , nearer 16.** Since  $15^2 = 225$  and  $16^2 = 256$ , and 250 is closer to 225.

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**Q9** **No.**  $2800 = 2^4 \times 5^2 \times 7$ ; the 7 has no pair, so it is not a perfect square.

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### Long answer

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**Q10** **105.**  $11025 = (3 \times 3)(5 \times 5)(7 \times 7)$ , so  $\sqrt{\phantom{x}} = 3 \times 5 \times 7 = 105$ .

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**Q11** **Divide by 6;  $\sqrt{\text{quotient}} = 6$ .**  $216 = 2^3 \times 3^3$ ; divide by  $2 \times 3 = 6 \rightarrow 36$ , and  $\sqrt{36} = 6$ .

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**Q12** **34 m.** Side =  $\sqrt{1156} = \sqrt{(2^2 \times 17^2)} = 2 \times 17 = 34$ .

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Total: 21 marks.